

EEE F313: ADVD Lab 5 Report

Pranav Chandra N. V.
2023AAPS0013

1 Introduction

This lab was the introductory lab to analog design in Cadence. Specifically, it involved the analysis of various parameters like g_m , I_D , V_{DS} , V_{GS} , and V_{TH} of a MOSFET. Analysis was covered for both a PMOS and an NMOS transistor, using the library provided by Mohali, pvh and nvh respectively.

2 NMOS Analysis

The NMOS transistor used for this analysis was the nvh transistor from the Mohali library. The first step was designing the circuit in Cadence Virtuoso. The circuit is shown in Figure ??.

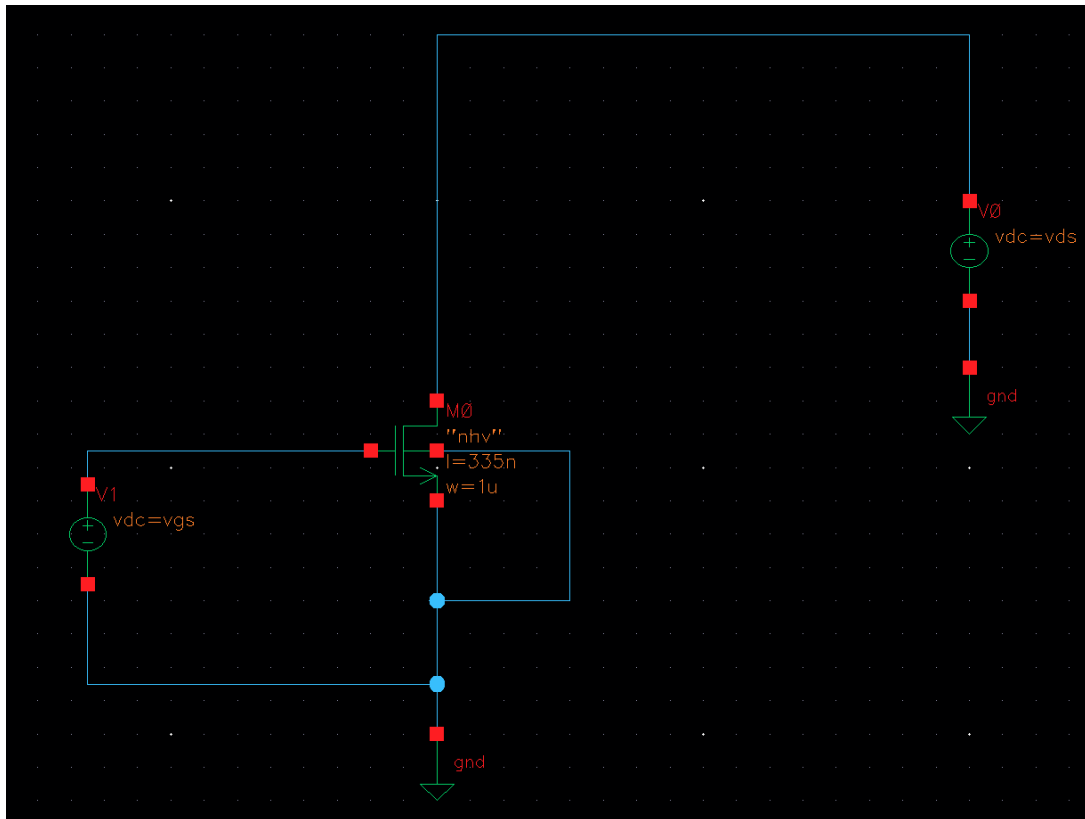
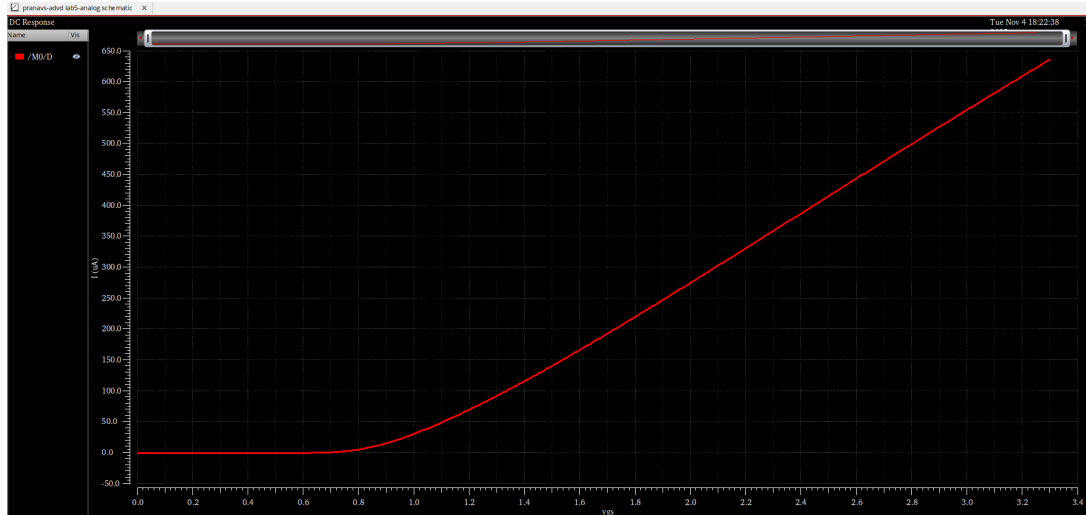


Figure 1: NMOS Analysis Circuit

Figure 2: I_D vs V_{GS} for NMOS Transistor

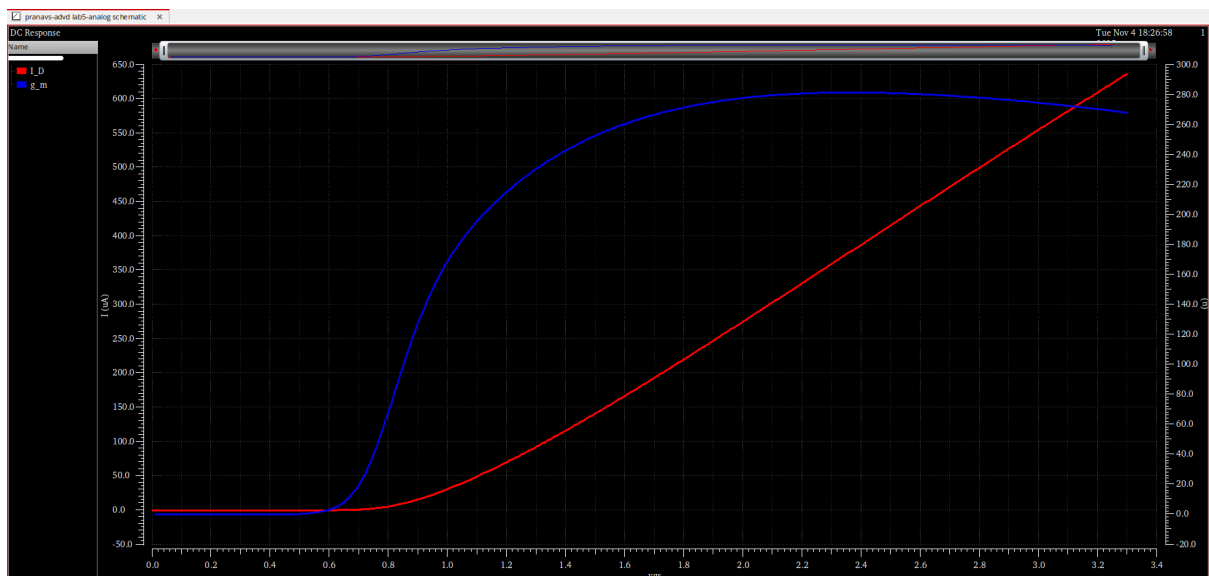
The next step was to perform a DC analysis of the circuit. The DC analysis was performed by sweeping the gate voltage (V_{GS}) from 0V to 3.3V.

The drain voltage (V_{DS}) was kept constant at 3.3V. The drain current (I_D) was measured for each value of V_{GS} . The results of the DC analysis are shown in Figure ??.

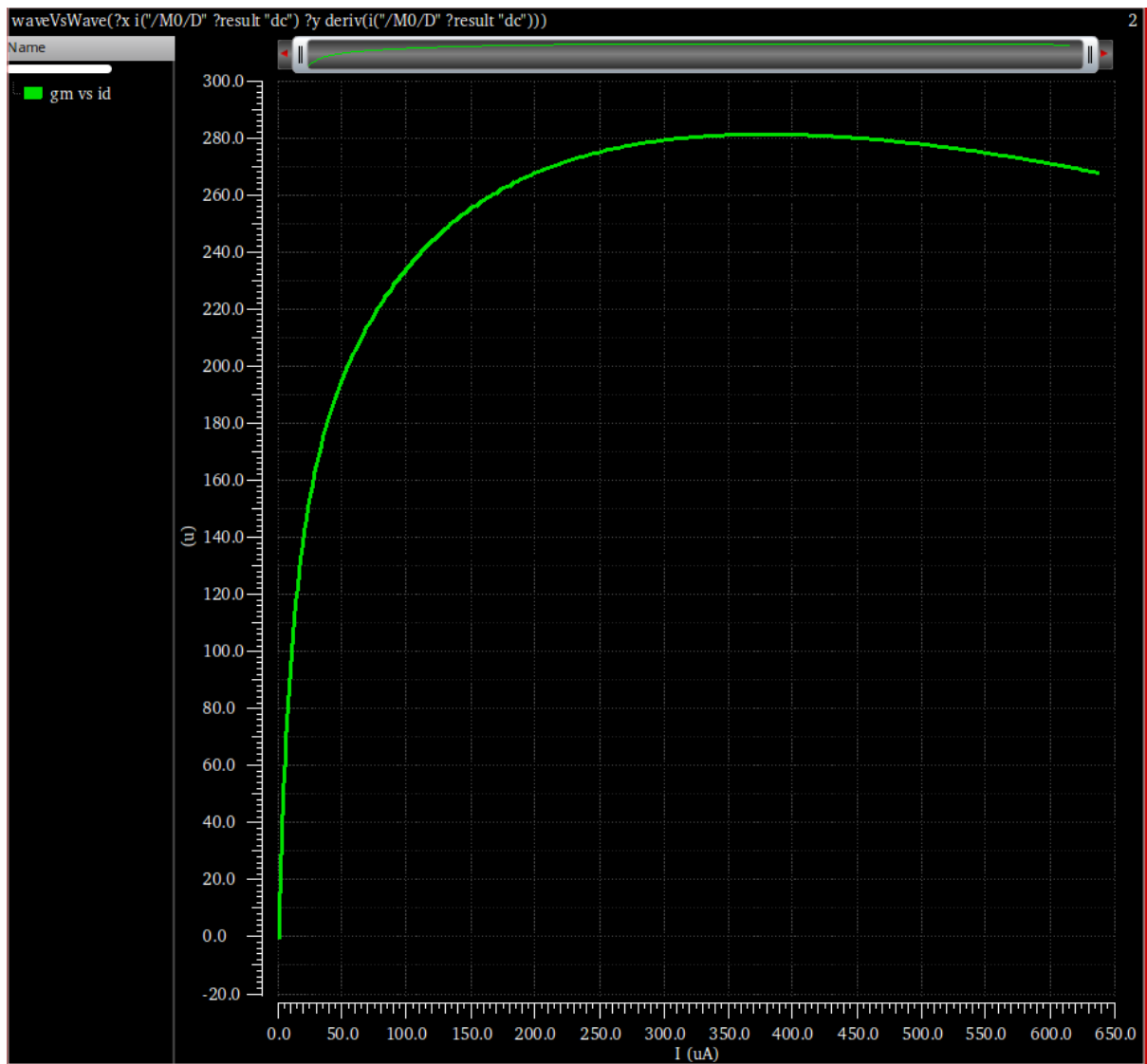
From here, g_m was the next obvious step. g_m was calculated using the formula:

$$g_m = \frac{dI_D}{dV_{GS}} \quad (1)$$

The derivative function in the calculator of Cadence was used in order to produce the g_m vs V_{GS} graph shown in Figure ??.

Figure 3: g_m vs V_{GS} (Blue) for NMOS Transistor

Now, the g_m vs I_D graph was plotted using the data from the previous two graphs. The results are shown in Figure ??. The graph calculator was used in Cadence in order to plot the requisite quantities against each other.

Figure 4: g_m vs I_D for NMOS Transistor

3 PMOS Analysis

The PMOS transistor used for this analysis was the pvh transistor from the Mohali library. The circuit designed in Cadence Virtuoso is shown in Figure ??.

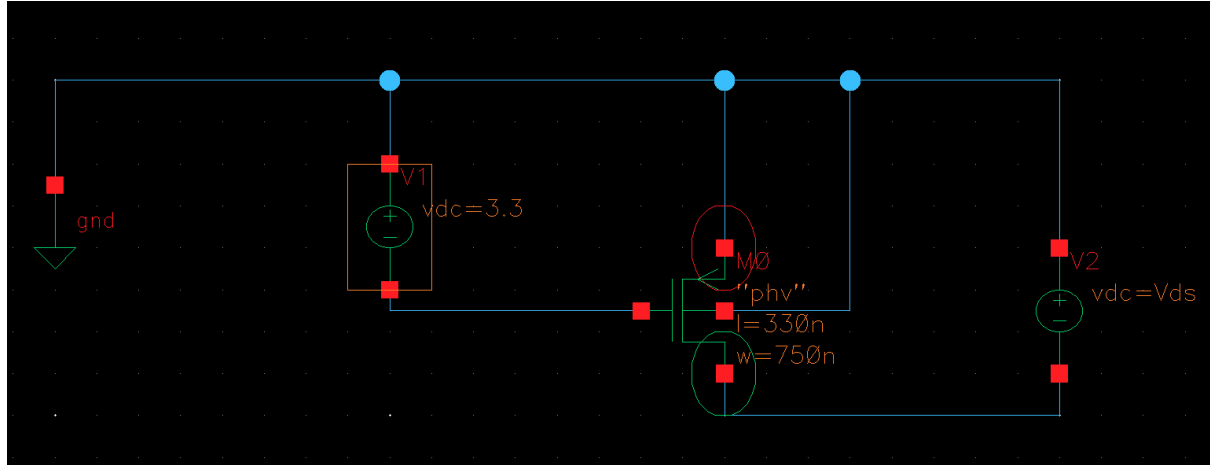


Figure 5: PMOS Analysis Circuit

The DC analysis was performed by sweeping the gate voltage (V_{GS}) from -3.3V to 0V. The drain voltage (V_{DS}) was kept constant at -3.3V. The drain current (I_D) was measured for each value of V_{GS} . The results of the DC analysis are shown in Figure ??.

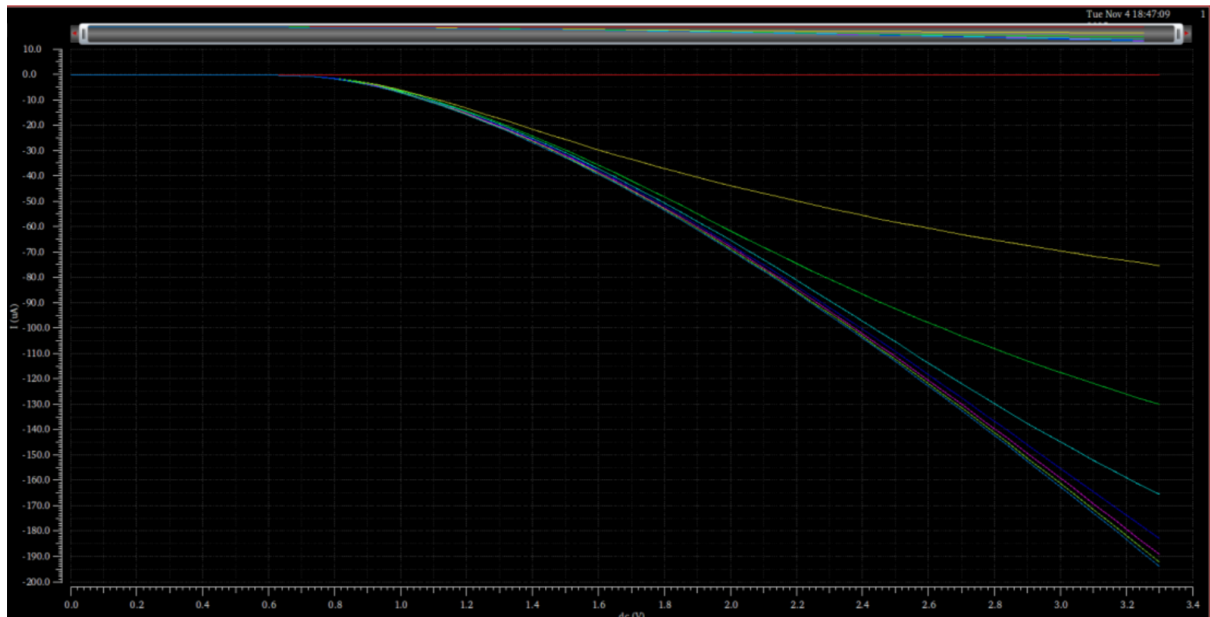
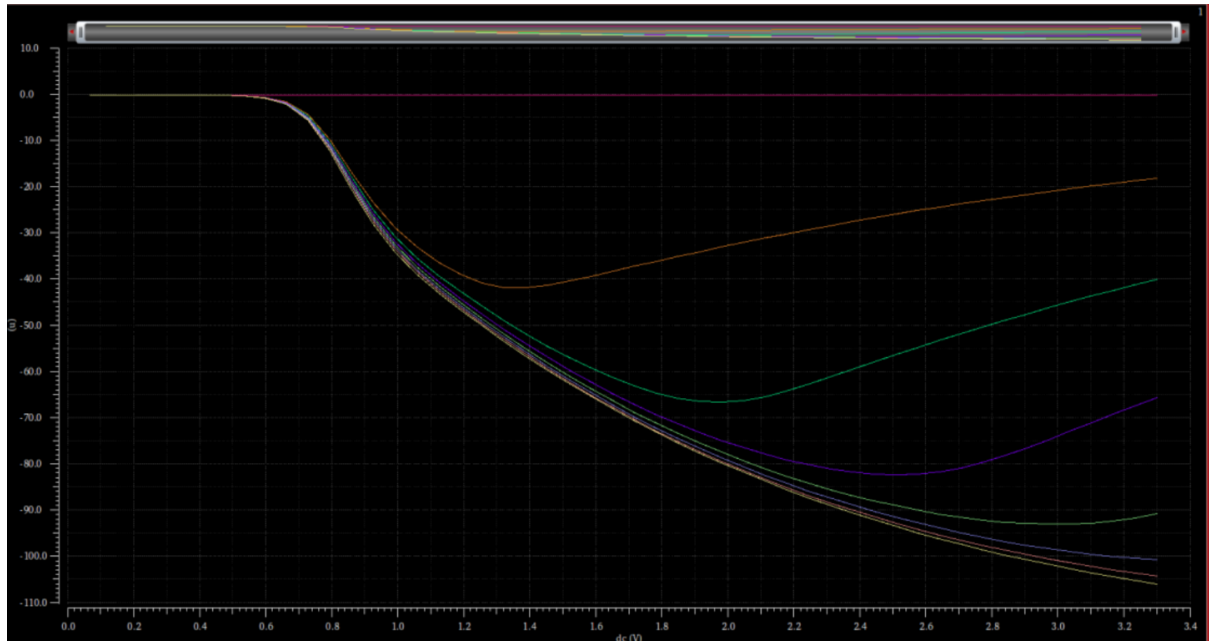


Figure 6: I_D vs V_{GS} for PMOS Transistor

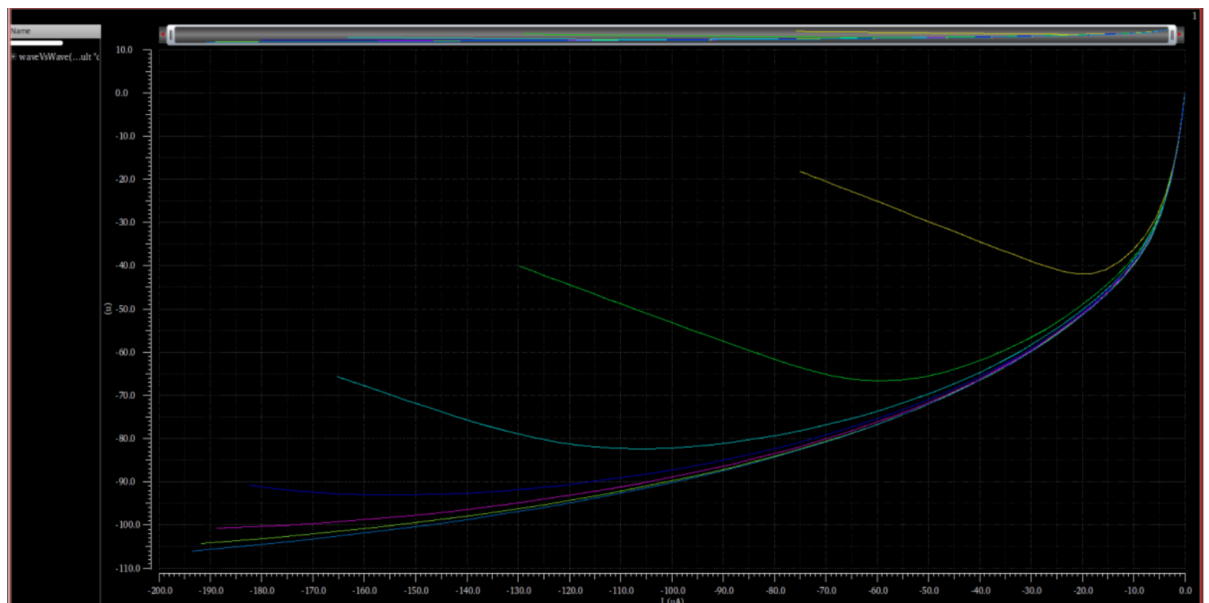
From here, g_m was calculated using the formula:

$$g_m = \frac{dI_D}{dV_{GS}} \quad (2)$$

The derivative function in the calculator of Cadence was used in order to produce the g_m vs V_{GS} graph shown in Figure ??.

Figure 7: g_m vs V_{GS} (Blue) for PMOS Transistor

Now, the g_m vs I_D graph was plotted using the data from the previous two graphs. The results are shown in Figure ???. The graph calculator was used in Cadence in order to plot the requisite quantities against each other.

Figure 8: g_m vs I_D for PMOS Transistor

4 Conclusion

In this lab, we successfully analyzed the characteristics of both NMOS and PMOS transistors using Cadence Virtuoso. We plotted the I_D vs V_{GS} curves and calculated the transconductance (g_m) for both types of transistors. The results obtained from the simulations were consistent with the theoretical expectations, demonstrating the fundamental behavior of MOSFETs in analog circuits. This lab provided a solid foundation for understanding transistor operation and will be beneficial for future analog design tasks.